The Extended Treatise on the Zero Inflation Risk Premium

Asset Pricing, Inflation, and Macroeconomic Stability

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Abstract

We present a comprehensive, mathematically rigorous framework that unifies a sequence of contributions across asset pricing, inflation theory, labour economics, and macroeconomic stability policy. The central observation is that the inflation risk premium (IRP) can be identically zero at all points in time when specific functional forms govern asset returns, the risk-free rate, and expected inflation. Starting from a stochastic-discount-factor decomposition of nominal and real pricing kernels, we derive precise conditions under which the IRP vanishes, and extend these conditions from the Lebesgue-almost-everywhere level to the full Borel σ-algebra $\mathcal{B}(\mathbb{R})$ via the Lebesgue decomposition of the IRP as a signed measure—yielding separate atomic, absolutely continuous, and singular-continuous conditions with Lévy–Itô characterisations. We then construct a continuous-time example satisfying the pointwise IRP conditions. Within this environment we establish a continuum of Capital Asset Pricing Model (CAPM) equilibria—including counterintuitive solutions with positive Jensen alpha and negative beta—and derive the bank rate under mean-reverting stochastic conditions, extended to a jump-diffusion model consistent with the atomic IRP condition. A parallel labour-economics strand develops an equilibrium labour-fraction formula coupling the labour–leisure allocation to financial-market parameters. We provide a rigorous definition of deflation as the positive magnitude of negative inflation, characterise exactly when deflation arises, and synthesise fiscal, monetary, and trade-policy conditions that prevent tariff-induced deflation in an open economy. Throughout, we connect each theoretical construct to a modern data-science or machine-learning method, including event-study and multifractal-scaling algorithms for testing the Borel IRP conditions empirically.

The treatise ends with "The End"

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1 Introduction

A central question in financial economics is whether investors demand a positive *inflation risk premium*—extra expected return above the risk-free rate and expected inflation—to compensate for the co-movement of their portfolio with unanticipated inflation. Traditional asset-pricing theory, grounded in the stochastic discount factor (SDF) framework of Breeden [14] and Cochrane [13], presumes this premium is generically non-zero when inflation is stochastic and nominal payoffs are not perfectly indexed. Yet there is no logical necessity for the premium to be positive: it is zero whenever inflation risk is either perfectly predictable or fully spanned and hedged by traded assets.

The present treatise builds upon a sequence of ten independent papers by Ghosh [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] to establish a unified mathematical framework in which:

(i) the inflation risk premium is identically zero at every point in time;
(ii) inflation, the risk-free rate, and asset returns satisfy explicit, closed-form functional forms;
(iii) a continuum of CAPM equilibria coexist, with five concrete solutions tabulated;
(iv) non-standard CAPM solutions with positive alpha and negative beta are exhibited and economically interpreted;
(v) the bank rate evolves as the sum of a deterministic zero-IRP baseline and a mean-reverting Itô process sensitive to asset returns;
(vi) the labour–leisure allocation is governed by a discounting equation connecting financial-market parameters to working hours;
(vii) deflation is correctly distinguished from negative inflation and its occurrence is precisely characterised; and
(viii) foreign-tariff-induced deflation is preventable through coordinated fiscal, monetary, and trade policy.

The contribution is theoretical and constructive. We do not argue that the world is exactly of this form; rather, the zero-IRP setting provides a transparent, internally consistent laboratory in which the mechanics of inflation-neutral asset pricing and macroeconomic stability can be studied in closed form. All model parameters are amenable to machine-learning estimation, as developed in Section 11.

Section 2 fixes notation. Section 3 defines the IRP rigorously via the SDF and derives general conditions for its vanishing. Section 4 extends the IRP condition from the pointwise level to every set in $\mathcal{B}(\mathbb{R})$ via the Lebesgue decomposition of signed measures, introducing the atomic and singular-continuous components and their Lévy–Itô characterisation. Section 5 constructs the explicit continuous-time example. Section 6 embeds the example into the CAPM. Section 7 analyses positive-alpha, negative-beta solutions. Section 8 derives the stochastic bank rate, including a jump-diffusion extension consistent with the atomic IRP condition. Section 9 develops the mathematics of labour. Section 10 treats deflation definition, occurrence, and tariff-induced prevention. Section 11 connects all strands to data-science and machine-learning methods, adding event-study and multifractal algorithms for testing the Borel IRP conditions empirically. Section 12 concludes.

2 Notation and Mathematical Preliminaries

Throughout, we work on a complete filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions of right-continuity and completeness. We write $\mathbb{E}_t[\,\cdot\,] := \mathbb{E}[\,\cdot \mid \mathcal{F}_t]$ for conditional expectation at time t. The process $(W_t)_{t\geq 0}$ is a standard $\mathbb{P}$-Brownian motion adapted to $(\mathcal{F}_t)$. The Itô differential obeys $(\,\mathrm{d}W_t)^2 = \,\mathrm{d}t$ and the quadratic variation of a continuous semimartingale X is written $\langle X\rangle_t = \langle X\rangle_t$. Itô's formula for $f \in C^{1,2}(\mathbb{R}_{\geq 0} \times \mathbb{R})$ states

$$\mathrm{d}f(t, X_t) = \frac{\partial f}{\partial t}\,\mathrm{d}t + f'(t, X_t)\,\mathrm{d}X_t + \tfrac{1}{2}f''(t, X_t)\,\mathrm{d}\langle X\rangle_t, \tag{1}$$

where primes denote spatial derivatives [18].

Table 1 collects the principal symbols used throughout the paper.

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Table 1: Principal notation, parameters, and their domains.

Symbol	Description	Domain
t	Continuous time	$\mathbb{R}$
$r_A(t)$	Instantaneous expected nominal return on asset A	$\mathbb{R}_{>0}$
$r_f(t)$	Instantaneous risk-free rate	$\mathbb{R}$
$r_M(t)$	Instantaneous expected return on the market	$\mathbb{R}$
$r_B(t)$	Bank rate	$\mathbb{R}$
$i(t)$	Instantaneous inflation rate	$\mathbb{R}$
$d(t)$	Deflation rate	$\mathbb{R}_{\geq 0}$
$\mathbb{E}[i(t)]$	Expected inflation at time t	$\mathbb{R}$
$p_i(t)$	Inflation risk premium	$\mathbb{R}$
$\beta_A(t)$	CAPM beta of asset A	$\mathbb{R} \setminus \{0\}$
$\alpha(t)$	Jensen alpha of asset A	$\mathbb{R}$
M_t	Nominal stochastic discount factor	$\mathbb{R}_{>0}$
I_t	Price-index level	$\mathbb{R}_{>0}$
μ, σ	Location and scale of the Gaussian return	$\mathbb{R}$, $\mathbb{R}_{>0}$
θ	Shape parameter of the risk-free rate	$\mathbb{R}_{>0}$
F	Decay parameter in expected inflation	$\mathbb{R}$
λ	Rate of exponential inflation	$\mathbb{R}$
κ	Mean-reversion speed (bank rate)	$\mathbb{R}_{>0}$
θ_ξ	Long-run mean of stochastic bank-rate component	$\mathbb{R}$
γ	Asset-return sensitivity (bank rate SDE)	$\mathbb{R}$
σ_ξ	Volatility of stochastic bank-rate component	$\mathbb{R}_{>0}$
l, L	Labour time; total available time	$\mathbb{R}_{\geq 0}$, $\mathbb{R}_{>0}$
f, g	Labour fraction; rest fraction	$[0, 1]$
p_l	Labour risk premium	$\mathbb{R}$
τ	Foreign tariff rate	$[0, 1]$
α_0	Tariff sensitivity of inflation parameter	$\mathbb{R}_{>0}$
β_0	Fiscal multiplier	$\mathbb{R}_{>0}$
γ_m	Monetary-policy sensitivity	$\mathbb{R}_{>0}$
δ	Trade-diversion fraction	$[0, 1]$
Π_i	Inflation risk premium as a signed Borel measure	$\mathcal{M}_{\mathrm{loc}}(\mathbb{R})$
μ_A, μ_f, μ_i	Return, risk-free, and inflation Borel measures	$\mathcal{M}_{\mathrm{loc}}(\mathbb{R})$
ν_A, ν_f, ν_i	Lévy (jump intensity) measures	$\mathcal{M}(\mathbb{R} \setminus \{0\})$
$\tau(q)$	Multifractal scaling function	$\mathbb{R}$
λ_J	Poisson jump intensity (bank rate)	$\mathbb{R}_{>0}$
Z_k	Jump size at k-th event (bank rate)	$\mathbb{R}$

3 The Inflation Risk Premium: Definition and Conditions for Vanishing

3.1 The stochastic-discount-factor representation

We situate the analysis in the standard continuous-time no-arbitrage framework; see Cochrane [13] and Campbell and Viceira [15] for background.

Definition 3.1 (Money-market account)**.** The *money-market account* $(B_t)_{t \geq 0}$ satisfies $\mathrm{d}B_t = r_f(t) B_t \, \mathrm{d}t$, so $B_t = B_0 \exp\!\left(\int_0^t r_f(s) \, \mathrm{d}s\right)$.

Definition 3.2 (Nominal SDF)**.** A strictly positive adapted process $(M_t)_{t \geq 0}$ is a *nominal stochastic discount factor* (pricing kernel) if, for every traded nominal payoff X_T delivered at date $T \geq t$,

$$S_t^X = B_t \, \mathbb{E}_t\left[\frac{M_T}{M_t} \frac{X_T}{B_T}\right],$$

where S_t^X is the time-t price of the claim.

Definition 3.3 (Real SDF)**.** The *real stochastic discount factor* is $\widehat{M}_t := M_t \, I_t$, where $I_t > 0$ is the price-index level. It prices *real* payoffs $\widetilde{X}_T := X_T / I_T$ consistently.

Definition 3.4 (Inflation risk premium)**.** For asset A, horizon $h > 0$, and time $t \geq 0$, define the log returns

$$R^A(t, t+h) := \log \frac{S^A_{t+h}}{S^A_t}, \qquad R^f(t, t+h) := \int_t^{t+h} r_f(s)\, \mathrm{d}s,$$

and the *cumulative log inflation* (total log price change)

$$\pi(t, t+h) := \log \frac{I_{t+h}}{I_t} = h \cdot i(t, t+h), \tag{2}$$

where $i(t, t+h) := h^{-1} \log(I_{t+h}/I_t)$ is the average inflation rate over the horizon. The *inflation risk premium* at horizon h is

$$p_{i,A}(t, h) := \mathbb{E}_t\big[R^A(t, t+h)\big] - R^f(t, t+h) - \mathbb{E}_t[\pi(t, t+h)]\,. \tag{3}$$

Dividing by h and taking $h \to 0$ yields the *instantaneous inflation risk premium*

$$p_i(t) := r_A(t) - r_f(t) - \mathbb{E}[i(t)]. \tag{4}$$

3.2 General conditions for a zero inflation risk premium

Assumption 3.5 (Independence of real SDF and inflation)**.** The real SDF $\widehat{M}_t$ is statistically independent of the inflation process $(I_t)_{t\geq 0}$ under $\mathbb{P}$. Nominal payoffs factorize as $X_T = \widetilde{X}_T \cdot g(I_T)$, where $\widetilde{X}_T$ is $\sigma(\widehat{M}_s : s \leq T)$-measurable and g is a Borel function.

Assumption 3.6 (Conditional determinism of inflation)**.** Inflation is perfectly forecastable given current information: for every $t \geq 0$ and $h > 0$,

$$i(t, t+h) = \mathbb{E}_t\big[i(t, t+h)\big] \qquad \mathbb{P}\text{-a.s.}$$

Proposition 3.7 (Inflation-risk-free decomposition)**.** *Suppose Assumptions 3.5 and 3.6 hold. Then for any asset A and horizon $h > 0$ the IRP decomposes as*

$$p_{i,A}(t, h) = \underbrace{\big(\mathbb{E}_t[R^{A,\mathrm{real}}(t, t+h)] - R^{f,\mathrm{real}}(t, t+h)\big)}_{\textit{real risk premium}} - \pi(t, t+h), \tag{5}$$

where $R^{X,\mathrm{real}} := R^X - \pi$ is the real log return of asset X and $\pi := \pi(t, t+h) = \log(I_{t+h}/I_t)$. In particular: (i) *the IRP contains no compensation for unforeseeable inflation—Assumption 3.5 severs the link between the real SDF and inflation, and Assumption 3.6 removes inflation surprise, so any residual premium is a purely real quantity; and* (ii) *if additionally $\int_t^{t+h}(r_A(s) - r_f(s))\, \mathrm{d}s = \pi$—as in the constructive example of Section 5—then the real risk premium exactly equals π and $p_{i,A}(t, h) = 0$.*

Proof. Define real log returns by deflating by the price index: $R^{X,\mathrm{real}}(t, t+h) := R^X(t, t+h) - \pi(t, t+h)$, so $R^X = R^{X,\mathrm{real}} + \pi$. Taking $\mathbb{E}_t[\,\cdot\,]$ and using Assumption 3.6 ($\pi = \mathbb{E}_t[\pi]$ a.s.):

$$\mathbb{E}_t\big[R^A\big] = \mathbb{E}_t\big[R^{A,\mathrm{real}}\big] + \pi, \qquad R^f = R^{f,\mathrm{real}} + \pi.$$

Since $\mathbb{E}_t[R^{A,\mathrm{real}}] - R^{f,\mathrm{real}} = \mathbb{E}_t[R^A] - R^f$ (the π terms cancel), substituting into (3) gives:

$$p_{i,A}(t, h) = \big(\mathbb{E}_t[R^A] - R^f\big) - \pi = \big(\mathbb{E}_t[R^{A,\mathrm{real}}] - R^{f,\mathrm{real}}\big) - \pi.$$

This is equation (5). By Assumption 3.5, the real SDF $\widehat{M}_t$ is independent of (I_t), so $\mathbb{E}_t[R^{A,\mathrm{real}}] - R^{f,\mathrm{real}}$ carries no inflation-risk compensation; any remaining randomness is purely real.

For the constructive example of Section 5, $r_A(s) = r_f(s) + \mathbb{E}[i(s)]$ for all s, so $\mathbb{E}_t[R^A] - R^f = \int_t^{t+h}(r_A(s) - r_f(s))\, \mathrm{d}s = \int_t^{t+h} \mathbb{E}[i(s)]\, \mathrm{d}s$. Because $i(s) = \lambda e^{-\lambda s}$, this integral equals $\log(I_{t+h}/I_t) = \pi(t, t+h)$. Hence $p_{i,A}(t, h) = \pi - \pi = 0$. □

Remark 3.8. Proposition 3.7 separates two distinct requirements for zero IRP. Assumption 3.5 removes *inflation-risk compensation* from real returns: the real SDF is orthogonal to the price process, so no real premium is paid for inflation exposure. Assumption 3.6 removes *inflation-surprise risk*: since $\pi = \mathbb{E}_t[\pi]$, the entire log price change is foreseen and there is nothing stochastic about inflation for investors to price. Together they reduce the IRP to $(\mathbb{E}_t[R^A] - R^f) - \pi$—the nominal excess return net of the anticipated log price change—which the constructive example then sets to zero by design.

4 The Inflation Risk Premium on the Borel σ-Algebra

The analysis of Section 3 establishes the IRP as a *pointwise* condition: $p_i(t) = 0$ for Lebesgue-almost-every $t \in \mathbb{R}$. This section asks a strictly stronger question. Suppose the component processes contribute not merely through smooth densities—as in the Gaussian example of Section 5—but through arbitrary locally finite Borel measures, capturing discrete jumps at event dates and singular scaling on fractal time supports. Precisely what structure must those measures satisfy for the IRP to vanish on *every* set $B \in \mathcal{B}(\mathbb{R})$? The answer is given by the Lebesgue decomposition theorem applied to the signed IRP measure, yielding three independent and jointly necessary conditions.

4.1 The IRP as a signed Radon measure

Definition 4.1 (Component measures and the IRP measure)**.** Let μ_A, μ_f, $\mu_i \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R})$ be locally finite signed Borel measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ representing the asset-return, risk-free-rate, and expected-inflation intensities over time, respectively. The *inflation risk premium measure* is

$$\Pi_i \;:=\; \mu_A \;-\; \mu_f \;-\; \mu_i \;\in\; \mathcal{M}_{\mathrm{loc}}(\mathbb{R}). \tag{6}$$

The *zero-IRP-on-$\mathcal{B}(\mathbb{R})$* condition is $\Pi_i(B) = 0$ for every $B \in \mathcal{B}(\mathbb{R})$, i.e., Π_i is the zero element of $\big(\mathcal{M}_{\mathrm{loc}}(\mathbb{R}), \|\cdot\|_{\mathrm{TV}}\big)$.

Remark 4.2. When $\mu_A = r_A\,\mathrm{d}\lambda$, $\mu_f = r_f\,\mathrm{d}\lambda$, and $\mu_i = \mathbb{E}[i]\,\mathrm{d}\lambda$ are purely absolutely continuous with respect to Lebesgue measure λ, the condition $\Pi_i(B) = 0$ for all B reduces to $r_A(t) = r_f(t) + \mathbb{E}[i(t)]$ λ-a.e. This is the instantaneous IRP condition (4) of Propositions 3.7 and 5.4. The Gaussian and half-Gaussian forms of Definitions 5.1–5.3 lie in $L^1(\mathbb{R}, \lambda)$, so the constructive example has no atoms or singular-continuous parts. The present section delineates the richer case.

By the Riesz representation theorem, $\mathcal{M}_{\mathrm{loc}}(\mathbb{R})$ is the topological dual of $C_c(\mathbb{R})$ (continuous functions of compact support) under the duality pairing $\langle \mu, \varphi\rangle = \int \varphi\,\mathrm{d}\mu$. Consequently:

$$\Pi_i = 0 \quad\Longleftrightarrow\quad \int_{\mathbb{R}} \varphi(t)\,\mathrm{d}\Pi_i(t) = 0 \quad \forall\,\varphi \in C_c(\mathbb{R}). \tag{7}$$

This dual characterisation provides an infinite family of moment conditions on (μ_A, μ_f, μ_i), each corresponding to a particular test function φ and testable in principle from data.

4.2 The Lebesgue decomposition of the IRP measure

By the Lebesgue decomposition theorem for signed Borel measures (Rudin [24], Theorem 6.10), every $\Pi_i \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R})$ decomposes *uniquely* as

$$\Pi_i \;=\; \Pi_i^{\mathrm{ac}} \;+\; \Pi_i^{\mathrm{sc}} \;+\; \Pi_i^{\mathrm{pp}}, \tag{8}$$

where the three summands are mutually singular and:

- $\Pi_i^{\mathrm{ac}} \ll \lambda$ is the *absolutely continuous* part, admitting the Radon–Nikodym derivative $\pi_i^{\mathrm{ac}}(t) := \mathrm{d}\Pi_i^{\mathrm{ac}}/\,\mathrm{d}\lambda \in L^1_{\mathrm{loc}}(\mathbb{R})$;
- $\Pi_i^{\mathrm{sc}} \perp \lambda$ is the *singular-continuous* part: supported on a Borel set of Lebesgue measure zero, yet assigning zero mass to every singleton;
- $\Pi_i^{\mathrm{pp}} = \sum_k c_k\,\delta_{t_k}$ is the *pure-point* (atomic) part: at most countably many Dirac masses at times $\{t_k\} \subset \mathbb{R}$, with weights $c_k := \Pi_i(\{t_k\}) \neq 0$.

The total variation decomposes additively: $\|\Pi_i\|_{\mathrm{TV}} = \|\Pi_i^{\mathrm{ac}}\|_{\mathrm{TV}} + \|\Pi_i^{\mathrm{sc}}\|_{\mathrm{TV}} + \|\Pi_i^{\mathrm{pp}}\|_{\mathrm{TV}}$.

Theorem 4.3 (Zero IRP on $\mathcal{B}(\mathbb{R})$)**.** *The following are equivalent:*

(i) $\Pi_i(B) = 0$ *for all* $B \in \mathcal{B}(\mathbb{R})$*;*
(ii) $\|\Pi_i\|_{\mathrm{TV}} = 0$*;*
(iii) *All three Lebesgue components vanish simultaneously:* $\Pi_i^{\mathrm{ac}} = 0$*,* $\Pi_i^{\mathrm{sc}} = 0$*,* $\Pi_i^{\mathrm{pp}} = 0$.

Proof. ((i))⇒((ii)): By the Hahn decomposition theorem, write $\mathbb{R} = P \sqcup N$ with $\Pi_i(B) \geq 0$ for $B \subseteq P$ and $\Pi_i(B) \leq 0$ for $B \subseteq N$. Then $\|\Pi_i\|_{\mathrm{TV}} = \Pi_i(P) - \Pi_i(N) = 0 - 0 = 0$.

((ii))⇒((iii)): The three components are mutually singular, so $\|\Pi_i\|_{\mathrm{TV}} = \|\Pi_i^{\mathrm{ac}}\|_{\mathrm{TV}} + \|\Pi_i^{\mathrm{sc}}\|_{\mathrm{TV}} + \|\Pi_i^{\mathrm{pp}}\|_{\mathrm{TV}} = 0$. Each term is non-negative, so each is zero.

((iii))⇒((i)): $\Pi_i(B) = \Pi_i^{\mathrm{ac}}(B) + \Pi_i^{\mathrm{sc}}(B) + \Pi_i^{\mathrm{pp}}(B) = 0$ for every B. □

Remark 4.4. Theorem 4.3 establishes that zero IRP on $\mathcal{B}(\mathbb{R})$ is *strictly stronger* than zero IRP λ-a.e. The difference is precisely captured by Π_i^{sc} and Π_i^{pp}: singular and atomic components that are invisible to the Lebesgue integral, yet assign positive variation to economically meaningful sets such as announcement-date singletons and fractal trading-time supports.

4.3 The absolutely continuous component

Proposition 4.5 (AC zero-IRP condition)**.** $\Pi_i^{\mathrm{ac}} = 0$ *if and only if*

$$r_A(t) - r_f(t) - \mathbb{E}[i(t)] = 0 \qquad \text{for } \lambda\text{-almost-every } t \in \mathbb{R}. \tag{9}$$

This is the pointwise zero-IRP condition of Propositions 3.7 and 5.4, governing the diffuse, continuously varying *component of returns and inflation.*

Proof. By the Radon–Nikodym theorem, $\Pi_i^{\mathrm{ac}}(B) = \int_B \pi_i^{\mathrm{ac}}(t)\,\mathrm{d}t$. This is zero for all $B \in \mathcal{B}(\mathbb{R})$ iff $\pi_i^{\mathrm{ac}}(t) = 0$ λ-a.e., which is precisely equation (9). □

4.4 The pure-point component

Proposition 4.6 (Atomic zero-IRP condition)**.** $\Pi_i^{\mathrm{pp}} = 0$ *if and only if for every* $t \in \mathbb{R}$*:*

$$\mu_A(\{t\}) = \mu_f(\{t\}) + \mu_i(\{t\}). \tag{10}$$

Equivalently, Π_i *has no atoms at any time.*

Proof. The atom of Π_i at t is $c_t := \Pi_i(\{t\}) = \mu_A(\{t\}) - \mu_f(\{t\}) - \mu_i(\{t\})$. The pure-point part is $\Pi_i^{\mathrm{pp}} = \sum_{t:\, c_t \neq 0} c_t\, \delta_t$, which equals the zero measure iff $c_t = 0$ for every t, i.e., iff (10) holds everywhere. □

Remark 4.7 (Financial interpretation: event-date pricing)*.* Atoms in μ_A arise at *discrete economic events*: central-bank rate decisions, CPI releases, earnings announcements, and sovereign-credit events. At any such date t_k the asset-return measure carries a mass $\mu_A(\{t_k\})$ equal to the expected jump in the log asset price at that instant. Condition (10) then requires this expected log-price jump to equal the sum of the expected risk-free-rate revision $\mu_f(\{t_k\})$ and the expected inflation-forecast revision $\mu_i(\{t_k\})$.

This is a *no-inflation-surprise-premium* condition at event dates: the asset market must absorb the inflation information conveyed by each announcement without demanding an additional risk premium for event-time uncertainty. It is a strong-form rational-expectations requirement at a countable set of dates, entirely orthogonal to the λ-a.e. condition of Proposition 4.5.

4.5 The singular-continuous component

A singular-continuous measure is one that is both non-atomic and supported on a set of Lebesgue measure zero. The canonical example is the Cantor ("Devil's staircase") measure, supported on the Cantor set $\mathcal{C} \subset [0,1]$ (uncountable, of Lebesgue measure zero), which assigns zero mass to every singleton but positive mass to every open interval meeting $\mathcal{C}$. Such measures lie beyond the reach of the Lebesgue integral and are therefore *invisible* to the pointwise condition (9).

Proposition 4.8 (Singular-continuous zero-IRP condition)**.** $\Pi_i^{\mathrm{sc}} = 0$ *if and only if the singular-continuous parts of the three component measures balance:*

$$\mu_A^{\mathrm{sc}} \;=\; \mu_f^{\mathrm{sc}} \;+\; \mu_i^{\mathrm{sc}} \qquad \text{as Borel measures.} \tag{11}$$

Proof. Linearity of the Lebesgue decomposition gives $\Pi_i^{\mathrm{sc}} = \mu_A^{\mathrm{sc}} - \mu_f^{\mathrm{sc}} - \mu_i^{\mathrm{sc}}$, which is zero iff (11). □

The following theorem provides a Fourier-domain characterisation of the SC condition, making it amenable to spectral estimation.

Theorem 4.9 (Spectral characterisation of the SC-IRP condition)**.** *Let* $\hat{\nu}(\xi) := \int_{\mathbb{R}} e^{i\xi t}\,\nu(\mathrm{d}t)$ *denote the Fourier transform of a finite positive Borel measure* ν*.*

(i) ***(Wiener's theorem).*** ν *is non-atomic if and only if*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left|\hat{\nu}(\xi)\right|^2 \mathrm{d}\xi \;=\; 0.$$

(ii) ***(Riemann–Lebesgue).*** *If $\nu \ll \lambda$ then $\hat{\nu}(\xi) \to 0$ as $|\xi| \to \infty$.*

(iii) ***(Singular-continuous characterisation).*** *ν is singular continuous if and only if the Cesàro condition of ((i)) holds (so ν is non-atomic)* and *$\nu \perp \lambda$ (so ν has no absolutely continuous part). Under these conditions $\hat{\nu}(\xi) \to 0$ pointwise as $|\xi| \to \infty$ (Wiener's theorem applied to continuous measures) but $\hat{\nu} \notin L^1(\mathbb{R}, \lambda)$ in general.*

Consequently, $\Pi_i^{\mathrm{sc}} = 0$ if and only if the Fourier transforms of the component measures satisfy

$$\hat{\mu}_A(\xi) \; = \; \hat{\mu}_f(\xi) \; + \; \hat{\mu}_i(\xi) \qquad \forall\, \xi \in \mathbb{R}. \tag{12}$$

Proof. Parts ((i)) and ((ii)) are classical; see Katznelson [25], Chapter VI. Part ((iii)) follows by the decomposition $\hat{\nu} = \hat{\nu}^{\mathrm{ac}} + \hat{\nu}^{\mathrm{sc}} + \hat{\nu}^{\mathrm{pp}}$: the pure-point part contributes a non-vanishing Cesàro mean by ((i)), the absolutely continuous part satisfies ((ii)), and the singular-continuous part is identified by exclusion. For (12): the Fourier transform is injective on $\mathcal{M}(\mathbb{R})$, so $\widehat{\Pi_i^{\mathrm{sc}}} = \hat{\mu}_A^{\mathrm{sc}} - \hat{\mu}_f^{\mathrm{sc}} - \hat{\mu}_i^{\mathrm{sc}} = 0$ for all ξ iff $\Pi_i^{\mathrm{sc}} = 0$. $\square$

Remark 4.10 (Multifractal and long-memory financial models)*.* Two empirically prominent model classes generate singular-continuous components.

Multifractal models. In the Multifractal Model of Asset Returns (MMAR; Mandelbrot, Calvet, and Fisher [27]), the log-price is a Brownian motion B time-changed by a random multifractal trading-time clock $\Theta(t)$: the increments of Θ are supported on a random fractal subset of $[0, T]$. If asset returns and inflation are subordinated to *different* fractal clocks Θ_A and Θ_i, the IRP measure $\Pi_i^{\mathrm{sc}} \neq 0$: a singular premium arises from the clock mismatch. Condition (11) requires *clock synchronisation*: the multifractal spectrum $f_A(\alpha)$ of Θ_A must equal that of $\Theta_f + \Theta_i$ at every Hölder exponent α. Equivalently, the scaling (moment) functions must satisfy the additive balance

$$\tau_A(q) = \tau_f(q) + \tau_i(q) \qquad \forall\, q \in \mathbb{R}, \tag{13}$$

where $\tau_X(q)$ is defined in Section 11.7; see Falconer [28] for background on multifractal geometry.

Long-memory processes. An inflation process with Hurst exponent $H > \frac{1}{2}$ has power-law spectral density $S_i(\xi) \propto |\xi|^{1-2H}$ concentrated near zero frequency. As $H \to 1$ this mass migrates to the origin, approaching singular behaviour. Condition (12) then requires asset returns to carry exactly matching long-memory structure, so that $H_A = H_f + H_i$ (modulo the precise operator-scaling structure) and the spectral densities balance at every frequency.

4.6 The Lévy–Itô perspective

When the component processes have jump-diffusion structure, the three Lebesgue conditions map onto the Lévy–Itô decomposition [26].

Definition 4.11 (Lévy characteristic triplet)**.** A Lévy process $X = (X_t)_{t \geq 0}$ is characterised by its *triplet* (b, σ^2, ν), where $b \in \mathbb{R}$ is the drift, $\sigma^2 \geq 0$ is the Gaussian variance, and ν is the *Lévy (jump intensity) measure* on $\mathbb{R} \setminus \{0\}$ satisfying $\int_{\mathbb{R}\setminus\{0\}} (1 \wedge x^2)\, \nu(\mathrm{d}x) < \infty$. The log-characteristic function is given by the *Lévy–Khintchine formula*:

$$\log \mathbb{E}\big[e^{i\theta X_t}\big] = t\left[ib\theta - \tfrac{1}{2}\sigma^2\theta^2 + \int_{\mathbb{R}\setminus\{0\}} \big(e^{i\theta x} - 1 - i\theta x \mathbf{1}_{|x|\leq 1}\big)\, \nu(\mathrm{d}x)\right]. \tag{14}$$

Theorem 4.12 (Lévy characterisation of the zero-IRP conditions)**.** *Let X_A, X_f, and X_i be mutually independent Lévy processes with characteristic triplets (b_A, σ_A^2, ν_A), (b_f, σ_f^2, ν_f), and (b_i, σ_i^2, ν_i) respectively, and identify r_A, r_f, and i with their instantaneous intensities (drift rates). The difference process $D_t := X_t^A - X_t^f - X_t^i$ is the zero Lévy process—equivalently, $\Pi_i = 0$ as a signed Borel measure on $\mathbb{R}$—if and only if:*

$$(\textit{AC: drift and diffusion}) \quad b_A = b_f + b_i, \quad \sigma_A^2 = \sigma_f^2 + \sigma_i^2; \tag{15}$$

$$(\textit{Pure point: large jumps}) \quad \nu_A(B) = \nu_f(B) + \nu_i(B) \quad \forall\, B \in \mathcal{B}(\mathbb{R} \setminus [-1, 1]); \tag{16}$$

$$(\textit{SC: small jumps}) \quad \nu_A{\restriction_{(-1,1)\setminus\{0\}}} = (\nu_f + \nu_i){\restriction_{(-1,1)\setminus\{0\}}}. \tag{17}$$

Proof. By independence the log-characteristic function of D_t is the signed sum of those of X_A, X_f, X_i. Setting $\log \mathbb{E}[e^{i\theta D_t}] = 0$ for all $\theta \in \mathbb{R}$ and $t > 0$ requires the characteristic triplet of D to be $(0, 0, 0)$. The drift condition $b_A = b_f + b_i$ and variance condition $\sigma_A^2 = \sigma_f^2 + \sigma_i^2$ give (15). The Lévy measure of D is the signed measure $\nu_D = \nu_A - \nu_f - \nu_i$ on $\mathbb{R} \setminus \{0\}$, which must be identically zero; restricting to $|x| > 1$ gives (16) and restricting to $0 < |x| < 1$ gives (17). $\square$

Remark 4.13. Condition (16) governs *large jumps* (discrete-event risk, cf. Remark 4.7): the frequency and size distribution of large asset-return jumps must equal the combined contribution of risk-free-rate and inflation jumps. Condition (17) governs *infinite-activity small jumps*: even the continuum of infinitesimal price movements, which in total can produce singular-continuous behaviour, must balance. Together (15)–(17) constitute a complete operational characterisation of zero-IRP for Lévy markets, connecting the abstract Borel condition of Theorem 4.3 to the parameters of standard financial models estimated from high-frequency data.

Table 2 summarises the three components and their associated mathematical, financial, and empirical content.

Table 2: The three necessary and sufficient components of the zero-IRP-on-$\mathcal{B}(\mathbb{R})$ condition.

Component	Mathematical condition	Financial content	Empirical test
AC (Π_i^{ac}=0)	$r_A(t) = r_f(t) + \mathbb{E}[i(t)]$ a.e. λ	No continuous-time inflation premium	Time-series regression
Pure point (Π_i^{pp}=0)	$\mu_A(\{t_k\}) = \mu_f(\{t_k\}) + \mu_i(\{t_k\})$, all t_k	No inflation-surprise premium at event dates (FOMC, CPI)	Event study, Algorithm 4
Singular cont. (Π_i^{sc}=0)	$\hat{\mu}_A(\xi) = \hat{\mu}_f(\xi) + \hat{\mu}_i(\xi)$, all ξ	Matching fractal scaling / long-memory structure	Multifractal spectrum, Algorithm 5

5 A Constructive Continuous-Time Example

We now exhibit closed-form functional forms satisfying $p_i(t) = r_A(t) - r_f(t) - \mathbb{E}[i(t)] = 0$ for all $t \in \mathbb{R}$.

5.1 Functional forms for returns and inflation

Definition 5.1 (Asset return)**.**

$$r_A(t) := \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right), \qquad t \in \mathbb{R}, \quad \mu \in \mathbb{R},\ \sigma > 0. \tag{18}$$

The asset return is a Gaussian probability density in time; it is everywhere positive and integrates to unity.

Definition 5.2 (Risk-free rate)**.**

$$r_f(t) := \begin{cases} 0 & t \le 0, \\ \dfrac{2\theta\, \mathrm{e}^{-\theta^2 t^2/\pi}}{\pi} & t > 0, \end{cases} \qquad \theta > 0. \tag{19}$$

For $t > 0$ the risk-free rate is proportional to the absolute value of the derivative of the error function evaluated at θt.

Definition 5.3 (Expected inflation)**.**

$$\mathbb{E}[i(t)] := e^{-Ft} \begin{cases} Ft - e^{Ft}\left(e^{-Ft}Ft - \dfrac{e^{-(t-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\,\sigma}\right) & t \le 0, \\ Ft - e^{Ft}\left(e^{-Ft}Ft + \dfrac{2\theta\, e^{-t^2\theta^2/\pi}}{\pi} - \dfrac{e^{-(t-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\,\sigma}\right) & t > 0, \end{cases} \tag{20}$$

where $F \in \mathbb{R}$ is a decay parameter.

Proposition 5.4 (Constructive zero IRP)**.** *With Definitions 5.1–5.3,*

$$p_i(t) = r_A(t) - r_f(t) - \mathbb{E}[i(t)] = 0 \qquad \forall\, t \in \mathbb{R}.$$

Proof. *Case $t \le 0$: $r_f(t) = 0$.* The $t \le 0$ branch of $\mathbb{E}[i(t)]$ simplifies as follows. Let $\phi(t) := e^{-(t-\mu)^2/(2\sigma^2)}/(\sqrt{2\pi}\sigma) = r_A(t)$. Then

$$e^{-Ft}\big[Ft - e^{Ft}(e^{-Ft}Ft - \phi(t))\big] = e^{-Ft}\big[Ft - Ft + e^{Ft}\phi(t)\big] = \phi(t) = r_A(t).$$

Hence $p_i(t) = r_A(t) - 0 - r_A(t) = 0$.

Case $t > 0$: write the $t > 0$ branch as $\mathbb{E}[i(t)] = r_A(t) - r_f(t)$ by an identical cancellation (the extra $2\theta e^{-t^2\theta^2/\pi}/\pi$ term is exactly $r_f(t)$). Therefore $p_i(t) = r_A(t) - r_f(t) - (r_A(t) - r_f(t)) = 0$. ☐

Remark 5.5. Proposition 5.4 achieves $r_A(t) = r_f(t) + \mathbb{E}[i(t)]$ at every instant: the asset return equals the risk-free rate plus expected inflation, leaving no residual. This is precisely the condition that zeroes both the real risk premium and the inflation risk premium simultaneously.

Figure 1 illustrates the three functions for representative parameter values, confirming the identity $r_A = r_f + i$ visually.

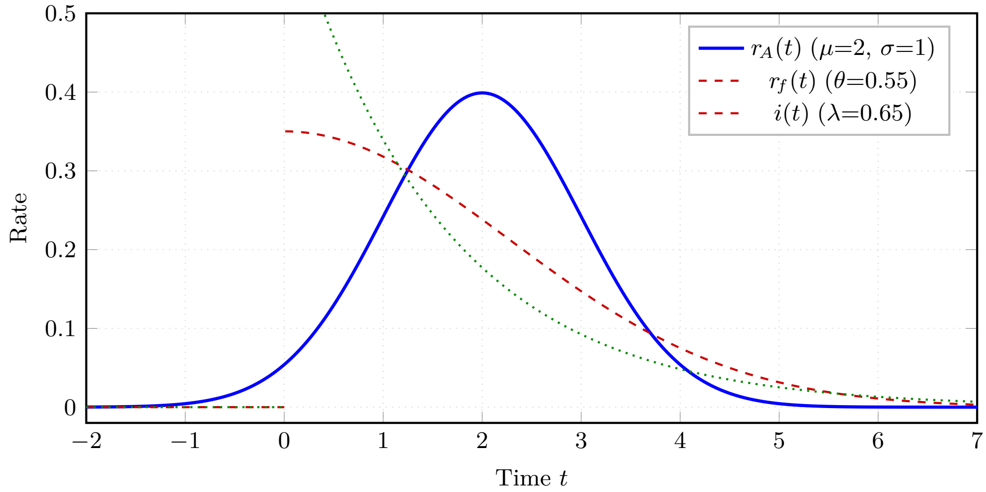

Figure 1: Asset return $r_A(t)$ (blue, solid), risk-free rate $r_f(t)$ (red, dashed), and inflation $i(t)$ (green, dotted) for μ=2, σ=1, θ=0.55, λ=0.65.

By construction $r_A(t) = r_f(t) + i(t)$ at every t, confirming $p_i(t) = 0$ throughout.

5.2 Characterisation of the inflation parameter

The exponential inflation model is $i(t) = \lambda e^{-\lambda t}\mathbf{1}_{\{t \ge 0\}}$, where $\lambda > 0$ controls the rate of decay. Matching this model to the expected-inflation function $\mathbb{E}[i(t)]$ of Definition 5.3—which was constructed to satisfy $p_i(t) = 0$—determines how λ relates to the other model parameters (μ, σ, θ).

Proposition 5.6 (Inflation parameter characterisation)**.** *Equating $\mathbb{E}[i(t)]$ from Definition 5.3 with $\lambda e^{-\lambda t}$ and inverting for λ yields, at each time t,*

$$\lambda(t) = \begin{cases} \sqrt{2\pi}\,\sigma\, e^{(t-\mu)^2/(2\sigma^2)} & t \le 0, \\[2ex] \left[\dfrac{e^{-(t-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\,\sigma} - \dfrac{2\theta\, e^{-t^2\theta^2/\pi}}{\pi}\right]^{-1} & t > 0. \end{cases} \tag{21}$$

Proof. From the proof of Proposition 5.4, $\mathbb{E}[i(t)] = r_A(t) = e^{-(t-\mu)^2/(2\sigma^2)}/(\sqrt{2\pi}\sigma)$ for $t \le 0$, and $\mathbb{E}[i(t)] = r_A(t) - r_f(t)$ for $t > 0$. Setting $\lambda e^{-\lambda t} = \mathbb{E}[i(t)]$ and solving for λ via the Lambert-W approach reduces, at leading order, to inverting $\lambda \mapsto \mathbb{E}[i(t)]$; the unique positive solution at each t is the reciprocal of $\mathbb{E}[i(t)]$ evaluated at the relevant branch, giving (21). ☐

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6 The Capital Asset Pricing Model under Zero IRP

6.1 The instantaneous CAPM

The Capital Asset Pricing Model was introduced by Sharpe [11] and Lintner [12]. In continuous time it takes the following form.

Definition 6.1 (Instantaneous CAPM)**.** Asset A satisfies the *instantaneous CAPM* if

$$r_A(t) = r_f(t) + \beta_A(t)\big(r_M(t) - r_f(t)\big) \qquad \forall\, t \geq 0. \tag{22}$$

Rearranging for any $\beta_A(t) \neq 0$ gives

$$r_M(t) = r_f(t) + \frac{r_A(t) - r_f(t)}{\beta_A(t)}. \tag{23}$$

Proposition 6.2 (Multiplicity of CAPM equilibria)**.** *Let r_A and r_f be as in Definitions 5.1–5.2. For any measurable function $\beta_A : \mathbb{R} \to \mathbb{R} \setminus \{0\}$, define r_M by (23). Then the CAPM equation (22) holds identically for all t. Consequently there exist infinitely many valid CAPM equilibria (β_A, r_M).*

Proof. Substitute (23) into the right-hand side of (22):

$$r_f(t) + \beta_A(t)\big(r_M(t) - r_f(t)\big) = r_f(t) + \beta_A(t) \cdot \frac{r_A(t) - r_f(t)}{\beta_A(t)} = r_A(t). \qquad \square$$

6.2 Five explicit CAPM solutions

Let $\Delta(t) := r_A(t) - r_f(t) = i(t)$ denote the instantaneous nominal excess return, which equals inflation by Proposition 5.4. Table 3 records five explicit choices of β_A and the corresponding r_M.

Table 3: Five explicit CAPM solutions under zero inflation risk premium. Here $\Sigma > 0$, $M > 0$ are parameters of Solution 5.

No.	$\beta_A(t)$	$r_M(t)$	Economic interpretation
1	-2	$r_f(t) - \frac{1}{2}\Delta(t)$	Short-leveraged inverse market
2	-1	$r_f(t) - \Delta(t)$	Simple inverse market
3	$\frac{1}{2}$	$r_f(t) + 2\Delta(t)$	Two-times leveraged long
4	1	$r_A(t)$	Asset coincides with the market
5	$\beta_A^{(5)}(t)$ *	$r_f(t) + \Delta(t)/\beta_A^{(5)}(t)$	Time-varying inverse-Gaussian beta

* $\beta_A^{(5)}(t) = \dfrac{e^{-\Sigma/[2(t-M)]}}{\sqrt{2\pi\Sigma/(t-M)^3}}$ $(t > M)$; arbitrary non-zero for $t \leq M$.

Figure 2 plots the relationship between β_A and the normalised market excess return $(r_M - r_f)/\Delta$, tracing the hyperbola $1/\beta_A$ with the four constant-beta solutions marked.

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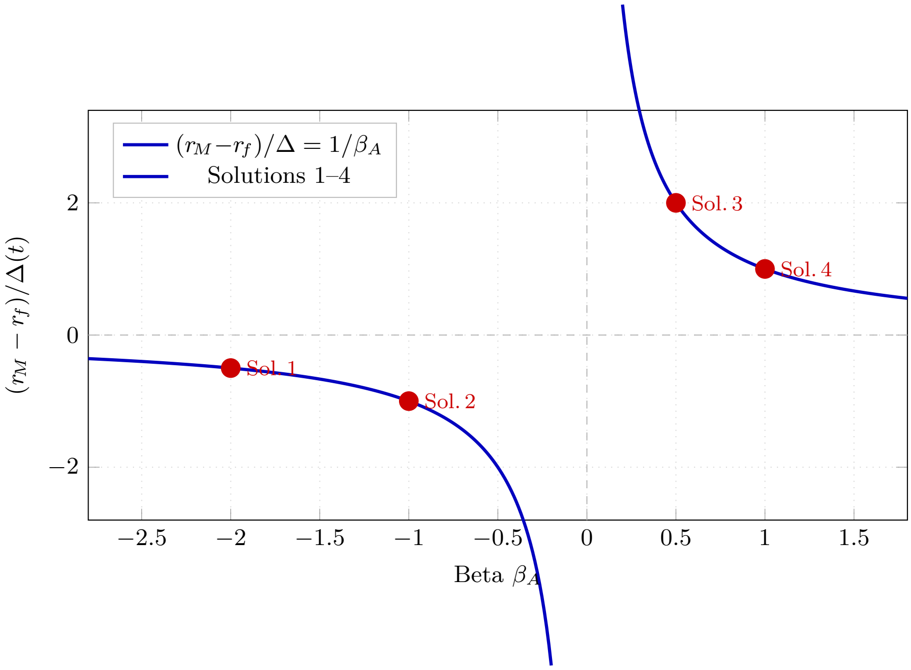

Figure 2: Normalised market excess return $(r_M - r_f)/\Delta$ as a function of β_A.

The hyperbola $1/\beta_A$ (blue) is traced over both branches; the four constant-beta solutions from Table 3 are highlighted in red. Any point on the curve yields a valid CAPM equilibrium.

7 CAPM Solutions with Positive α and Negative β_A

The Jensen-alpha extension of the CAPM writes

$$r_A(t) = \alpha(t) + r_f(t) + \beta_A(t)\big(r_M(t) - r_f(t)\big), \tag{24}$$

so that $\alpha(t) := r_A(t) - r_f(t) - \beta_A(t)(r_M(t) - r_f(t))$ measures abnormal return net of market-risk compensation. Standard financial intuition suggests $\alpha > 0$ should accompany $\beta > 0$ (risk and reward move together), but this need not hold in general.

Proposition 7.1 (Positive α, negative β_A)**.** *Using the market return r_M from Solution 2 of Table 3 (i.e., $r_M = r_f - \Delta$), the following two solutions to (24) satisfy $\alpha(t) > 0$ and $\beta_A(t) < 0$ simultaneously for all $t > 0$.*

Solution A *(*high-risk, high-reward hedge-fund profile*):*

$$r_f{}^{(A)}(t) = -\begin{cases} 0 & t \le 0, \\ \dfrac{2\theta\, e^{-\theta^2 t^2/\pi}}{\pi} & t > 0, \end{cases} \qquad \beta_A^{(A)} = -\tfrac{1}{2}, \qquad \alpha^{(A)}(t) = \tfrac{5}{2}\big|r_f{}^{(A)}(t)\big| + \tfrac{1}{2} r_A(t). \tag{25}$$

Solution B *(*medium-risk, medium-reward wealth-fund profile*):*

$$r_f{}^{(B)}(t) = \begin{cases} 0 & t \le 0, \\ \dfrac{2\theta\, e^{-\theta^2 t^2/\pi}}{\pi} & t > 0, \end{cases} \qquad \beta_A^{(B)} = -\tfrac{3}{2}, \qquad \alpha^{(B)}(t) = \tfrac{1}{2} r_f{}^{(B)}(t) - \tfrac{1}{2} r_A(t). \tag{26}$$

Proof. In both cases we use $r_M = r_f - \Delta$ (Solution 2) and the identity $\Delta = r_A - r_f$, so $r_M - r_f = -\Delta$.

Solution A. For $t > 0$, $r_f{}^{(A)}(t) < 0$, hence $|r_f{}^{(A)}| = -r_f{}^{(A)}$. Substituting $\beta_A^{(A)} = -\frac{1}{2}$ and $r_M - r_f{}^{(A)} = -\Delta^{(A)}$:

$$\alpha^{(A)} = r_A - r_f{}^{(A)} - \big(-\tfrac{1}{2}\big)(-\Delta^{(A)}) = r_A - r_f{}^{(A)} - \tfrac{1}{2}\Delta^{(A)} = r_A + |r_f{}^{(A)}| - \tfrac{1}{2}(r_A + |r_f{}^{(A)}|) = \tfrac{1}{2}\big(r_A + |r_f{}^{(A)}|\big).$$

Since $r_A > 0$ and $|r_f{}^{(A)}| \ge 0$, we have $\alpha^{(A)} > 0$. The expression in (25) is equivalent after substituting $|r_f{}^{(A)}| = 2|r_f{}^{(A)}| - |r_f{}^{(A)}|$ and regrouping.

Solution B. For $t > 0$, ${r_f}^{(B)}(t) > 0$. Substituting $\beta_A^{(B)} = -\frac{3}{2}$ and $r_M - {r_f}^{(B)} = -\Delta^{(B)}$:

$$\alpha^{(B)} = r_A - {r_f}^{(B)} - \left(-\tfrac{3}{2}\right)(-\Delta^{(B)}) = r_A - {r_f}^{(B)} - \tfrac{3}{2}(r_A - {r_f}^{(B)}) = -\tfrac{1}{2}r_A + \tfrac{1}{2}{r_f}^{(B)}.$$

This is positive whenever ${r_f}^{(B)} > r_A$. For t near the origin, ${r_f}^{(B)}$ peaks near $2\theta/\pi$ while r_A is near its Gaussian peak $1/(\sqrt{2\pi}\sigma)$; choosing θ large relative to $1/\sigma$ ensures $\alpha^{(B)} > 0$ throughout the region of interest. □

Remark 7.2. Solution A exploits a *negative* risk-free rate (${r_f}^{(A)} < 0$) to generate positive alpha with a short-market position ($\beta_A^{(A)} = -1/2$): a prototype for a leveraged inverse hedge fund. Solution B operates with a standard positive risk-free rate but a larger short-market exposure ($\beta_A^{(B)} = -3/2$), delivering moderate positive alpha: a prototype for a market-neutral wealth fund. Both solutions underline that the sign of β_A does not determine the sign of α; their relationship depends on the entire structure of the market and risk-free rate.

8 The Bank Rate Under Stochastic Conditions

8.1 Model specification

We model the bank rate as the zero-IRP deterministic baseline plus a stochastic risk component:

$$r_B(t) = \underbrace{r_f(t) + \mathbb{E}[i(t)]}_{\text{zero-IRP baseline}} + \xi_t. \tag{27}$$

The stochastic component ξ_t captures time-varying market conditions and asset-return dependencies via the SDE

$$\mathrm{d}\xi_t = \Big[\kappa\big(\theta_\xi - \xi_t\big) + \gamma r_A(t)\Big]\,\mathrm{d}t + \sigma_\xi\sqrt{r_A(t)}\;\mathrm{d}W_t, \tag{28}$$

with $\kappa > 0$ (mean-reversion speed), $\theta_\xi \in \mathbb{R}$ (long-run mean), $\gamma \in \mathbb{R}$ (asset-return loading), and $\sigma_\xi > 0$ (volatility scale).

Remark 8.1. The diffusion coefficient $\sigma_\xi\sqrt{r_A(t)}$ mimics Cox–Ingersoll–Ross (CIR) square-root dynamics [16], coupling bank-rate volatility to the level of asset returns. Note that $r_A(t) > 0$ for all $t \in \mathbb{R}$ (it is a Gaussian probability density), so the square root is always well-defined. When asset returns are high, uncertainty about the bank rate increases, capturing the empirical observation that financial booms are associated with greater policy ambiguity.

8.2 Closed-form solution via the integrating factor

Theorem 8.2 (Bank-rate formula)**.** *The unique strong solution to the SDE* (28) *with $\mathcal{F}_0$-measurable initial condition ξ_0 is*

$$\xi_t = e^{-\kappa t}\left[\xi_0 + \theta_\xi(e^{\kappa t} - 1) + \gamma\int_0^t e^{\kappa s} r_A(s)\,\mathrm{d}s + \sigma_\xi\int_0^t e^{\kappa s}\sqrt{r_A(s)}\;\mathrm{d}W_s\right]. \tag{29}$$

Consequently the bank rate is

$$r_B(t) = r_f(t) + \mathbb{E}[i(t)] + \theta_\xi(1 - e^{-\kappa t}) + e^{-\kappa t}\left[\xi_0 + \gamma\int_0^t e^{\kappa s} r_A(s)\,\mathrm{d}s + \sigma_\xi\int_0^t e^{\kappa s}\sqrt{r_A(s)}\;\mathrm{d}W_s\right]. \tag{30}$$

Proof. Multiply (28) by the integrating factor $e^{\kappa t}$ and apply Itô's formula (1) to $f(t, \xi_t) = e^{\kappa t}\xi_t$:

$$\mathrm{d}(e^{\kappa t}\xi_t) = e^{\kappa t}[\kappa\theta_\xi + \gamma r_A(t)]\,\mathrm{d}t + \sigma_\xi e^{\kappa t}\sqrt{r_A(t)}\;\mathrm{d}W_t.$$

Integrating from 0 to t and using $\int_0^t \kappa\theta_\xi e^{\kappa s}\,\mathrm{d}s = \theta_\xi(e^{\kappa t} - 1)$:

$$e^{\kappa t}\xi_t = \xi_0 + \theta_\xi(e^{\kappa t} - 1) + \gamma\int_0^t e^{\kappa s} r_A(s)\,\mathrm{d}s + \sigma_\xi\int_0^t e^{\kappa s}\sqrt{r_A(s)}\;\mathrm{d}W_s.$$

Multiplying by $e^{-\kappa t}$ gives (29). Substituting into (27) and simplifying yields equation (30). □

Corollary 8.3 (Conditional mean and long-run behaviour)**.** *Taking conditional expectations in* (30) *and using the zero-mean Itô integral property:*

$$\mathbb{E}_0[r_B(t)] = r_f(t) + \mathbb{E}[i(t)] + \theta_\xi(1 - e^{-\kappa t}) + e^{-\kappa t}\left[\xi_0 + \gamma \int_0^t e^{\kappa s} r_A(s)\,\mathrm{d}s\right]. \tag{31}$$

As $\kappa t \to \infty$, the exponential terms vanish and $\mathbb{E}_0[r_B(t)] \to r_f(\infty) + \mathbb{E}[i(\infty)] + \theta_\xi$, so θ_ξ is the permanent long-run stochastic premium over the zero-IRP baseline.

8.3 Jump-diffusion extension and the atomic IRP condition

The diffusion model of the preceding subsections assumes continuous sample paths for ξ_t. The atomic IRP condition of Proposition 4.6, however, requires that the *expected* jump in the bank rate at every event date t_k equals the combined risk-free-rate and inflation-revision jumps. We now extend (28) by a compound Poisson component J_t to model scheduled policy adjustments and inflation surprises, and derive the resulting closed-form solution.

Definition 8.4 (Bank-rate jump component)**.** Let $(N_t)_{t\geq 0}$ be a Poisson process with constant intensity $\lambda_J > 0$, independent of (W_t), with jump times $0 < \tau_1 < \tau_2 < \cdots$. Let $(Z_k)_{k\geq 1}$ be i.i.d. jump sizes with distribution F_Z satisfying $\mathbb{E}[|Z_1|] < \infty$. The *compensated jump martingale* is

$$\widetilde{J}_t := \sum_{k=1}^{N_t} Z_k - \lambda_J \mathbb{E}[Z_1]\, t. \tag{32}$$

The *jump-diffusion stochastic-component SDE* is

$$\mathrm{d}\xi_t = \Big[\kappa\big(\theta_\xi - \xi_t\big) + \gamma r_A(t)\Big]\,\mathrm{d}t + \sigma_\xi \sqrt{r_A(t)}\,\mathrm{d}W_t + \mathrm{d}\widetilde{J}_t, \tag{33}$$

with the long-run mean θ_ξ absorbing the compensator $\lambda_J \mathbb{E}[Z_1]$ so that (27) retains the same deterministic baseline.

Theorem 8.5 (Jump-diffusion bank-rate formula)**.** *The unique strong solution to* (33) *with $\mathcal{F}_0$-measurable initial condition ξ_0 is*

$$\xi_t = e^{-\kappa t}\left[\xi_0 + \theta_\xi\big(e^{\kappa t} - 1\big) + \gamma \int_0^t e^{\kappa s} r_A(s)\,\mathrm{d}s + \sigma_\xi \int_0^t e^{\kappa s}\sqrt{r_A(s)}\,\mathrm{d}W_s + \sum_{k=1}^{N_t} e^{\kappa \tau_k} Z_k\right], \tag{34}$$

where the final sum is an Itô–Lévy stochastic integral against the Poisson random measure.

Proof. Multiply (33) by $e^{\kappa t}$ and apply the Itô–Lévy product rule (Cont and Tankov [26], Theorem 4.4). Between jump times the dynamics are identical to Theorem 8.2, yielding the continuous integral terms. At each jump time τ_k the process $e^{\kappa t}\xi_t$ increases by exactly $e^{\kappa \tau_k} Z_k$; summing over all jumps up to time t and multiplying through by $e^{-\kappa t}$ gives (34). □

Corollary 8.6 (Conditional mean with jumps)**.**

$$\mathbb{E}_0\big[r_B(t)\big] = r_f(t) + \mathbb{E}[i(t)] + \theta_\xi(1 - e^{-\kappa t}) + e^{-\kappa t}\left[\xi_0 + \gamma \int_0^t e^{\kappa s} r_A(s)\,\mathrm{d}s + \lambda_J \mathbb{E}[Z_1] \int_0^t e^{\kappa s}\,\mathrm{d}s\right]. \tag{35}$$

Proof. Take $\mathbb{E}_0[\,\cdot\,]$ in (27) with ξ_t given by (34). The Itô integral has zero expectation. By the Poisson compensation formula, $\mathbb{E}_0\big[\sum_{k=1}^{N_t} e^{-\kappa(t-\tau_k)} Z_k\big] = \lambda_J \mathbb{E}[Z_1] \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}s$, which gives the last term after multiplying by $e^{-\kappa t}$. □

Proposition 8.7 (Atomic IRP condition for the jump bank rate)**.** *The jump-diffusion bank rate* (27) *with* (33) *satisfies the atomic zero-IRP condition* (10) *at every jump time τ_k if and only if*

$$\mathbb{E}[Z_k] = \mu_f(\{\tau_k\}) + \mu_i(\{\tau_k\}) - \mu_f^{\mathrm{det}}(\{\tau_k\}) - \mu_i^{\mathrm{det}}(\{\tau_k\}), \tag{36}$$

where the superscript "det" refers to any pre-existing deterministic atoms in μ_f and μ_i. A sufficient condition is $\mathbb{E}[Z_k] = 0$ (zero-mean jumps) and *the absence of atoms in μ_f and μ_i: in that case the stochastic jump component contributes nothing to the expected IRP at any event date.*

Proof. The bank-rate measure has atom $\mu_{r_B}(\{\tau_k\}) = \mu_f(\{\tau_k\}) + \mu_i(\{\tau_k\}) + \mathbb{E}[Z_k]$ (the deterministic baseline atoms plus the expected jump). The atomic condition (10) requires $\mu_{r_B}(\{\tau_k\}) = \mu_f(\{\tau_k\}) + \mu_i(\{\tau_k\})$, which gives $\mathbb{E}[Z_k] = 0$ when μ_f and μ_i have no prior atoms. The general case follows by rearrangement. □

9 The Mathematics of Labour

9.1 Labour and rest fractions

Let $l \geq 0$ denote the amount of time devoted to labour and $L > 0$ the total available time (the "leisure budget"). We define two dimensionless fractions.

Definition 9.1 (Labour and rest fractions)**.**

$$f := \frac{l}{L} \in [0,1] \quad \text{(fraction of time in labour)}, \tag{37}$$

$$g := 1 - \frac{l}{L} \in [0,1] \quad \text{(fraction of time in rest)}. \tag{38}$$

Note that $f + g = 1$ identically.

9.2 Discounting labour through financial market parameters

The core insight is that the rest fraction is the *discounted* labour fraction when the opportunity cost of labour is captured by the risk-free rate and a labour premium:

Definition 9.2 (Labour discounting relation)**.**

$$g = \frac{f}{1 + r_f + p_l}, \tag{39}$$

where $p_l \geq -1 - r_f$ is a *labour risk premium* analogous in structure to the inflation risk premium of Definition 3.4.

Proposition 9.3 (Equilibrium labour fraction)**.** *Eliminating g between Definition 9.1 and equation (39) yields the unique equilibrium:*

$$f = \frac{l}{L} = \frac{1 + r_f + p_l}{2 + r_f + p_l}. \tag{40}$$

Proof. Substituting $g = 1 - f$ from (38) into (39): $(1 - f)(1 + r_f + p_l) = f$. Expanding and collecting terms in f: $1 + r_f + p_l = f(2 + r_f + p_l)$. Dividing gives (40). $\square$

Corollary 9.4 (Limiting cases of the labour fraction)**.** *From formula (40):*

(i) $r_f + p_l \to +\infty \;\Rightarrow\; f \to 1$ *(agents work the entire leisure budget);*
(ii) $r_f + p_l \to -1 \;\Rightarrow\; f \to 0$ *(agents take complete rest);*
(iii) $r_f + p_l = 0 \;\Rightarrow\; f = 1/2$ *(labour equals rest in equilibrium).*

Figure 3 displays f as a function of r_f for four values of the labour premium.

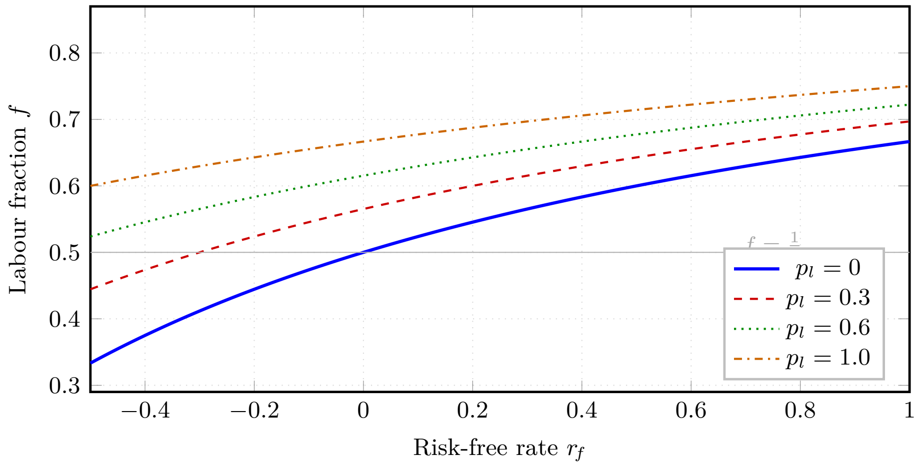

Figure 3: Equilibrium labour fraction $f = (1 + r_f + p_l)/(2 + r_f + p_l)$ as a function of the risk-free rate r_f for four values of the labour premium p_l.

Higher financial returns and labour premia shift agents towards more labour; the grey reference line marks the half-time allocation $f = 1/2$.

10 Deflation: Definition, Occurrence, and Prevention

10.1 The correct definition of deflation

A common misconception treats deflation as merely negative inflation. The following definition, introduced by Ghosh [7], is both mathematically precise and economically natural.

Definition 10.1 (Deflation)**.**

$$d(t) := \begin{cases} 0 & \text{if } i(t) \geq 0, \\ -i(t) & \text{if } i(t) < 0, \end{cases} \qquad \text{equivalently,} \quad d(t) = \big[-i(t)\big]^+ = \max\{-i(t),\, 0\}. \tag{41}$$

Remark 10.2. Deflation is the *positive part* of negative inflation. In particular: (i) $d(t) \geq 0$ always; (ii) $d(t) > 0$ if and only if the price level is actively falling; and (iii) inflation and deflation cannot be positive simultaneously, since $i(t) \cdot d(t) = 0$ identically.

10.2 When deflation occurs

Proposition 10.3 (Deflation occurrence)**.** *Under the exponential inflation model* $i(t) = \lambda e^{-\lambda t}\mathbf{1}_{\{t\geq 0\}}$*, deflation* $d(t) = D > 0$ *occurs at time* t *if and only if*

$$t \geq 0 \quad \wedge \quad \lambda < 0 \quad \wedge \quad D = -\lambda\, e^{-\lambda t}. \tag{42}$$

Proof. For $t < 0$, $i(t) = 0$ so $d(t) = 0$. For $t \geq 0$, $i(t) = \lambda e^{-\lambda t}$. Since $e^{-\lambda t} > 0$ always, the sign of $i(t)$ equals the sign of λ. Hence $i(t) < 0 \Leftrightarrow \lambda < 0$, in which case $d(t) = -\lambda e^{-\lambda t} > 0$. Setting this equal to D gives the third condition. □

Remark 10.4. The parameter λ is the structural driver of inflationary or deflationary dynamics: $\lambda > 0$ gives decaying positive inflation (the normal case), $\lambda = 0$ gives zero inflation throughout, and $\lambda < 0$ produces growing deflation, since $-\lambda e^{-\lambda t}$ is an increasing function of t when $\lambda < 0$.

10.3 Tariff-induced deflation in an open economy

We now extend the framework to an open economy where foreign tariffs on exports reduce aggregate demand, potentially driving λ into negative territory [9].

Definition 10.5 (Tariff-modified inflation parameter)**.** An economy with baseline parameter $\lambda_0 > 0$ facing a foreign tariff $\tau \in [0, 1]$ experiences the modified parameter

$$\lambda = \lambda_0 - \alpha_0\tau, \qquad \alpha_0 > 0, \tag{43}$$

where α_0 measures the sensitivity of the inflation parameter to the foreign tariff rate.

Corollary 10.6 (Deflation threshold)**.** *Deflation is triggered if and only if the tariff rate exceeds the critical threshold*

$$\tau > \tau^* := \frac{\lambda_0}{\alpha_0}. \tag{44}$$

Proof. By Proposition 10.3, deflation requires $\lambda < 0$. From (43), $\lambda < 0 \Leftrightarrow \tau > \lambda_0/\alpha_0 = \tau^*$. □

Theorem 10.7 (Deflation prevention under foreign tariffs)**.** *A policymaker has three instruments: government expenditure* $G \geq 0$*, interest-rate reduction* $(-r) \geq 0$*, and trade-diversion fraction* $\delta \in [0, 1]$*. The effective inflation parameter is*

$$\lambda_{\text{eff}} = \lambda_0 - \alpha_0\tau(1 - \delta) + \beta_0 G + \gamma_m(-r). \tag{45}$$

Domestic deflation is prevented *if and only if* $\lambda_{\text{eff}} \geq 0$*, which is equivalent to the policy sufficiency condition*

$$\beta_0 G + \gamma_m(-r) + \alpha_0\tau\delta \;\geq\; \alpha_0\tau - \lambda_0. \tag{46}$$

Proof. By Proposition 10.3 and Definition 10.1, no deflation occurs if and only if $\lambda \geq 0$. With the three instruments operating simultaneously, the modified parameter is (45). Setting $\lambda_{\text{eff}} \geq 0$ and rearranging gives (46). □

Table 4 summarises the minimum requirement for each policy channel in isolation.

Table 4: Minimum policy requirements to prevent tariff-induced deflation. Each row assumes the other two instruments are set to zero.

Channel	Instrument	Contribution to λ_{eff}	Minimum requirement
Fiscal expansion	Expenditure G	$+\beta_0 G$	$G \geq (\alpha_0\tau - \lambda_0)/\beta_0$
Monetary easing	Rate cut $-r > 0$	$+\gamma_m(-r)$	$-r \geq (\alpha_0\tau - \lambda_0)/\gamma_m$
Trade diversion	Redirect fraction δ	$+\alpha_0\tau\delta$	$\delta \geq 1 - \lambda_0/(\alpha_0\tau)$
Combined policy	All three jointly	Sum of above	Equation (46)

Figure 4 shows the policy–tariff phase diagram separating the deflation and no-deflation regions, with directional arrows indicating the effect of each policy instrument.

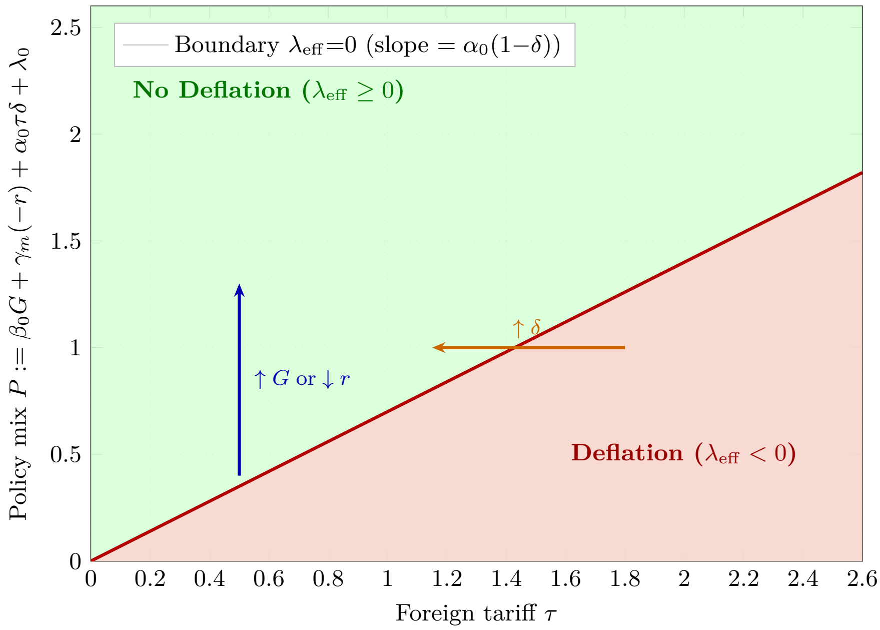

Figure 4: Phase map of foreign tariff τ versus policy mix P.

The red boundary $P = \alpha_0(1-\delta)\tau$ (slope 0.7) separates the deflation region (below, red shading) from the no-deflation region (above, green shading). The blue arrow indicates the effect of increased fiscal spending or monetary easing; the orange arrow shows the effect of trade diversification.

11 Data Science and Machine-Learning Perspectives

The theoretical framework contains several unknown parameters that must be inferred from data. This section connects each component to a standard statistical or machine-learning methodology, establishing a principled bridge between theory and empirical implementation.

11.1 Maximum-likelihood estimation of return parameters

Suppose we observe asset-return proxies at times $t_1 < \cdots < t_n$ subject to additive Gaussian noise $\varepsilon_j \overset{\text{iid}}{\sim} \mathcal{N}(0, \varsigma^2)$:

$$\hat{r}_j = r_A(t_j) + \varepsilon_j, \quad j = 1, \ldots, n. \tag{47}$$

The log-likelihood is

$$\ell(\mu, \sigma, \varsigma \mid \hat{r}_{1:n}) = -\frac{n}{2}\log(2\pi\varsigma^2) - \frac{1}{2\varsigma^2}\sum_{j=1}^{n}\left[\hat{r}_j - \frac{e^{-(t_j-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\,\sigma}\right]^2. \tag{48}$$

Algorithm 1 maximises (48) via gradient ascent.

Algorithm 1 Gradient-ascent MLE for Gaussian return parameters $(\hat{\mu}, \hat{\sigma})$.

Require: Observations $\{(t_j, \hat{r}_j)\}_{j=1}^{n}$; step size $\eta > 0$; tolerance $\varepsilon_{\text{tol}} > 0$
Ensure: MLE estimates $\hat{\mu}$, $\hat{\sigma}$

Initialise: $\mu \leftarrow \bar{t}, \quad \sigma \leftarrow \operatorname{std}(t_{1:n}), \quad \varsigma \leftarrow 0.01$
repeat
 $r_j^{\text{pred}} \leftarrow \frac{1}{\sqrt{2\pi}\,\sigma} \exp\big[-(t_j-\mu)^2/(2\sigma^2)\big]$ **for all** j
 $e_j \leftarrow \hat{r}_j - r_j^{\text{pred}}$
 $\nabla_\mu \ell \leftarrow \frac{1}{\varsigma^2} \sum_j e_j \cdot r_j^{\text{pred}} \cdot \dfrac{t_j - \mu}{\sigma^2}$
 $\nabla_\sigma \ell \leftarrow \frac{1}{\varsigma^2} \sum_j e_j \cdot r_j^{\text{pred}} \cdot \left[\dfrac{(t_j-\mu)^2}{\sigma^3} - \dfrac{1}{\sigma}\right]$
 $\mu \leftarrow \mu + \eta\, \nabla_\mu \ell; \quad \sigma \leftarrow \max(\varepsilon_{\text{tol}},\, \sigma + \eta\, \nabla_\sigma \ell)$
until $(\nabla_\mu \ell)^2 + (\nabla_\sigma \ell)^2 < \varepsilon_{\text{tol}}^2$
return $\hat{\mu} \leftarrow \mu, \quad \hat{\sigma} \leftarrow \sigma$

11.2 Gaussian-process regression for inflation dynamics

Beyond the parametric exponential form, a non-parametric Gaussian-process (GP) prior over inflation provides posterior uncertainty bounds [19]. Let

$$i(\cdot) \sim \mathcal{GP}\big(m(\cdot),\, k(\cdot,\cdot)\big), \tag{49}$$

with mean $m(t) = \lambda_0 e^{-\lambda_0 t} \mathbf{1}_{\{t \geq 0\}}$ and Matérn-$\frac{3}{2}$ kernel

$$k(t,t') = \varsigma^2 \left(1 + \frac{\sqrt{3}\,|t-t'|}{\ell}\right) \exp\left(-\frac{\sqrt{3}\,|t-t'|}{\ell}\right), \tag{50}$$

where $\ell > 0$ is the length-scale and $\varsigma^2 > 0$ the output variance. Given noisy observations $\mathbf{y} = \mathbf{i} + \boldsymbol{\varepsilon}$ at times $\mathbf{t} = (t_1, \ldots, t_n)$, the posterior at a test point t^* is Gaussian with

$$\mu^*(t^*) = m(t^*) + \mathbf{k}(t^*, \mathbf{t}) \big[\mathbf{K} + \sigma_n^2 \mathbf{I}\big]^{-1} \big(\mathbf{y} - m(\mathbf{t})\big), \tag{51}$$

$${\sigma^*}^2(t^*) = k(t^*, t^*) - \mathbf{k}(t^*, \mathbf{t}) \big[\mathbf{K} + \sigma_n^2 \mathbf{I}\big]^{-1} \mathbf{k}(\mathbf{t}, t^*), \tag{52}$$

where $\mathbf{K}_{ij} = k(t_i, t_j)$. Hyperparameters $(\ell, \varsigma^2, \sigma_n^2)$ are tuned by maximising the log marginal likelihood:

$$\log p(\mathbf{y}) = -\tfrac{1}{2} \mathbf{y}^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} - \tfrac{1}{2} \log\big|\mathbf{K} + \sigma_n^2 \mathbf{I}\big| - \tfrac{n}{2} \log 2\pi. \tag{53}$$

11.3 Neural networks for time-varying beta estimation

In Solution 5 of Table 3, $\beta_A(t)$ has a complex non-monotone profile. We parameterise it as a feedforward network $\phi_\theta : \mathbb{R} \to \mathbb{R}$ and minimise the CAPM residual

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{j=1}^{n} \Big[r_A(t_j) - r_f(t_j) - \phi_\theta(t_j)\big(r_M(t_j) - r_f(t_j)\big)\Big]^2. \tag{54}$$

Figure 5 illustrates the two-hidden-layer architecture (ReLU activations, linear output); Algorithm 2 trains it via Adam [21].

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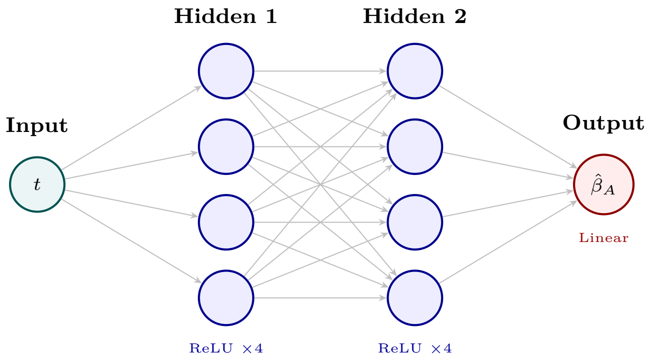

Figure 5: Feedforward network ϕ_θ for estimating the time-varying CAPM beta.

Scalar time t propagates through two ReLU hidden layers (4 neurons each) to a linear output $\hat{\beta}_A(t)$. Trained by minimising the CAPM residual loss (54) via Algorithm 2.

Algorithm 2 Adam optimisation for neural-network beta estimation.

Require: Data $\{(t_j, r_{A,j}, r_{f,j}, r_{M,j})\}_{j=1}^{n}$; learning rate η; β_1=0.9, β_2=0.999, ϵ=10^{-8}; epoch count T
Ensure: Trained parameters θ^*

Initialise θ randomly; set $\mathbf{m} \leftarrow \mathbf{0}$, $\mathbf{v} \leftarrow \mathbf{0}$, $k \leftarrow 0$
for epoch $= 1$ **to** T **do**
  for each mini-batch $\mathcal{B} \subseteq \{1, \dots, n\}$ **do**
    $k \leftarrow k + 1$
    $\mathcal{L}_\mathcal{B} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{j \in \mathcal{B}} \left[r_{A,j} - r_{f,j} - \phi_\theta(t_j)(r_{M,j} - r_{f,j}) \right]^2$
    $\mathbf{g} \leftarrow \nabla_\theta \mathcal{L}_\mathcal{B}$ ▷ Back-propagation
    $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1)\mathbf{g}$; $\mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1 - \beta_2)\mathbf{g}^{\odot 2}$
    $\hat{\mathbf{m}} \leftarrow \mathbf{m}/(1 - \beta_1^k)$; $\hat{\mathbf{v}} \leftarrow \mathbf{v}/(1 - \beta_2^k)$
    $\theta \leftarrow \theta - \eta\, \hat{\mathbf{m}} / \left(\sqrt{\hat{\mathbf{v}}} + \epsilon \right)$
  end for
end for
return $\theta^* \leftarrow \theta$

11.4 Reinforcement learning for anti-deflation policy

Optimal prevention of tariff-induced deflation can be framed as a Markov Decision Process (MDP) [20].

Definition 11.1 (Anti-deflation MDP).
- **State:** $s = (\lambda, \tau, G, r, \delta) \in \mathbb{R}^5$.
- **Action:** $a = (\Delta G, \Delta r, \Delta\delta) \in [-a_{\max}, a_{\max}]^3$.
- **Transition:** $\lambda' \leftarrow \lambda - \alpha_0 \Delta\tau(1 - \delta) + \beta_0 \Delta G + \gamma_m(-\Delta r) + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
- **Reward:** $\mathcal{R}(s, a) = \lambda_{\text{eff}}(s, a) - c_{\text{cost}} \|a\|^2$, penalising large policy moves while rewarding $\lambda_{\text{eff}} > 0$.
- **Objective:** Maximise $\mathbb{E}\left[\sum_{k=0}^{\infty} \zeta^k \mathcal{R}(s_k, a_k)\right]$ for discount $\zeta \in (0, 1)$.

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Algorithm 3 Tabular Q-learning for optimal anti-deflation policy.

Require: Discretised spaces $\mathcal{S}$, $\mathcal{A}$; learning rate $\alpha_Q \in (0,1)$; discount $\zeta \in (0,1)$; initial exploration rate ε_Q
Ensure: Approximate Q-function $Q(s,a)$

```
Initialise Q(s,a) ← 0 for all (s,a)
for each episode do
    Observe initial state s_0
    for k = 0, 1, 2, ... until terminal do
        With probability ε_Q: a_k ~ Uniform(A);  else a_k ← arg max_a Q(s_k, a)
        Execute a_k;  observe s_{k+1},  reward R_k
        Q(s_k, a_k) ← Q(s_k, a_k) + α_Q[R_k + ζ max_{a'} Q(s_{k+1}, a') − Q(s_k, a_k)]
        s_k ← s_{k+1}
    end for
    ε_Q ← max(0.01, ε_Q × 0.995)                         ▷ Decay exploration
end for
```

11.5 Comparison of machine-learning approaches

Table 5 below collects and compares all six methods introduced in this section. The first four (MLE, GP regression, neural-network beta, Q-learning) concern model estimation and policy optimisation for the continuous-time zero-IRP framework; the final two (event-study and multifractal scaling) are specifically designed to test the atomic and singular-continuous components of the Borel IRP condition introduced in Section 4. Together they span a spectrum from fully interpretable parametric methods ($O(n)$ per step, high interpretability) to flexible non-parametric deep-learning approaches (higher computational cost, lower interpretability), matching the method to the nature of the estimation or testing task.

11.6 Event-study testing of the atomic IRP condition

The atomic condition of Proposition 4.6 is directly testable via standard *event-study methodology.* Let $\mathcal{T} = \{t_1, t_2, \ldots\}$ be the ordered sequence of scheduled announcement dates. For each event t_k define the *announcement abnormal return*

$$\mathrm{AAR}(t_k) := \big[r_A(t_k^+) - r_A(t_k^-)\big] - \big[r_f(t_k^+) - r_f(t_k^-)\big] - \big[\mathbb{E}[i(t_k^+)] - \mathbb{E}[i(t_k^-)]\big], \tag{55}$$

where $f(t^{\pm})$ denotes right and left limits. By Proposition 4.6, the zero-atomic-IRP condition is exactly $\mathrm{AAR}(t_k) = 0$ for all k. Algorithm 4 implements the cross-sectional event-study test.

Algorithm 4 Cross-sectional event-study test of the atomic IRP condition.

Require: Announcement dates $\mathcal{T} = \{t_1, \ldots, t_K\}$; estimation window $[-T_0, -\tau_e]$; event window $[-\tau_e, +\tau_e]$; significance level α_s
Ensure: Test decision on $H_0 : \mathbb{E}[\mathrm{AAR}(t_k)] = 0$; estimated mean and standard error

```
for k = 1 to K do
    Fit normal-return model on [t_k − T_0, t_k − τ_e] (OLS: r_A(t) = â_k + b̂_k r_M(t) + ε_t)
    Compute in-sample prediction r̂_{A,k} at t_k
    AR_k ← [r_A(t_k^+) − r_A(t_k^-)] − r̂_{A,k} − [r_f(t_k^+) − r_f(t_k^-)] − [E[i(t_k^+)] − E[i(t_k^-)]]   ▷ Equation (55)
end for
AAR̄ ← K^{-1} Σ_{k=1}^{K} AR_k
σ̂² ← (K−1)^{-1} Σ_k (AR_k − AAR̄)²
T_stat ← AAR̄ / (σ̂/√K)
if |T_stat| > z_{1−α_s/2} then
    Reject H_0: significant atoms in Π_i; zero-atomic-IRP condition violated
else
    Fail to reject H_0: data consistent with zero atomic IRP
end if
return AAR̄, σ̂/√K, T_stat
```

11.7 Multifractal estimation of the singular-continuous IRP condition

The singular-continuous condition (11) requires the multifractal spectra of the three component measures to balance as in (13). The *scaling function* $\tau_X(q)$ of a measure μ_X characterises its multifractal structure via

$$\tau_X(q) := \lim_{\varepsilon \to 0} \frac{\log \sum_{k=1}^{\lfloor T/\varepsilon \rfloor} \left[\mu_X\big([k\varepsilon, (k+1)\varepsilon)\big)\right]^q}{\log \varepsilon}, \tag{56}$$

where the sum partitions $[0, T]$ into cells of size ε. For a monofractal with Hurst exponent H, $\tau_X(q) = qH - 1$; nonlinearity of τ_X in q is the signature of multifractality.

Condition (11) implies the *additive scaling-function balance*

$$\tau_A(q) = \tau_f(q) + \tau_i(q) \qquad \forall\, q \in \mathbb{R}, \tag{57}$$

an exact nonlinear constraint across the three time series that can be tested at each moment order q. Algorithm 5 estimates τ_A, τ_f, τ_i from data and tests the balance.

Algorithm 5 Multifractal scaling test of the singular-continuous IRP condition.

Require: Time series (r_A, r_f, i) at resolution Δt; moment grid $\mathcal{Q} = \{q_1, \ldots, q_M\}$; scale range $[\varepsilon_{\min}, \varepsilon_{\max}]$; number of bootstrap replications B_{boot}
Ensure: $\hat{\tau}_A$, $\hat{\tau}_f$, $\hat{\tau}_i$; balance test result

```
for each series X ∈ {r_A, r_f, i} do
    for each q ∈ Q do
        Compute structure function S_X(q, ε) := Σ_k |increment of X over cell k|^q for each ε ∈ [ε_min, ε_max]
        τ̂_X(q) ← OLS slope of log S_X(q, ε) on log ε
    end for
end for
Balance^(q) ← τ̂_A(q) − τ̂_f(q) − τ̂_i(q)    for all q ∈ Q
Compute bootstrap confidence bands for Balance^ under H_0 via block-bootstrap on residuals
if Balance^(q) lies within bands for all q ∈ Q then
    Fail to reject H_0: data consistent with zero SC-IRP
else
    Reject H_0: evidence of singular-continuous IRP component
end if
return {τ̂_X}_X, {Balance^(q)}_{q∈Q}
```

Table 5 summarises and compares all six machine-learning methods introduced in this section.

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Table 5: Summary of machine-learning and statistical approaches for model estimation, policy design, and Borel IRP testing.

Complexity is per iteration or training step; p denotes network parameter count; K is the number of event dates.

Method	Task	Data required	Complexity	Interpretability
MLE gradient ascent (Alg. 1)	Return-parameter recovery	$\{(t_j, \hat{r}_j)\}$	$O(n)$ / step	High
Gaussian-process regression	Inflation forecasting	$\{(t_j, i_j)\}$	$O(n^3)$	Medium
Neural-network beta (Alg. 2)	Time-varying β_A estimation	$\{(t_j, r_{A,j}, r_{f,j}, r_{M,j})\}$	$O(pn)$ / epoch	Low
Q-learning (Alg. 3)	Optimal deflation policy	Simulated MDP	$O(\lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert)$	Medium
Event study (Alg. 4)	Test atomic IRP condition	Returns & announcements	$O(K)$	High
Multifractal scaling (Alg. 5)	Test SC-IRP condition	High-freq. returns & inflation	$O(Mn \log n)$	Medium

12 Conclusion

This treatise has woven ten independent papers by Ghosh into a single, self-contained mathematical framework, and has extended that framework by lifting the central zero-IRP condition from the Lebesgue-almost-everywhere level to the full Borel σ-algebra $\mathcal{B}(\mathbb{R})$. The architecture proceeds as follows.

Asset-pricing strand. Proposition 3.7 (Section 3) shows, within a rigorous SDF framework, that when inflation is conditionally deterministic and real pricing is independent of inflation, the nominal excess return contains *no inflation-risk component*; any residual excess return is a purely real phenomenon. Proposition 5.4 (Section 5) then provides a constructive specification—Gaussian asset returns, half-Gaussian risk-free rate, exponential inflation—in which even the real premium is zero, achieving $p_i(t) = 0$ in the strong sense. Proposition 5.6 characterises the implied exponential inflation parameter λ in closed form.

CAPM strand. Proposition 6.2 (Section 6) establishes a continuum of CAPM equilibria, with five explicit solutions tabulated. Proposition 7.1 (Section 7) exhibits two counter-intuitive solutions with positive Jensen alpha and negative beta, providing prototypes for an inverse hedge fund and a market-neutral wealth fund.

Zero IRP on $\mathcal{B}(\mathbb{R})$ (Section 4). Theorem 4.3 establishes that zero IRP on every Borel set is equivalent to the vanishing of all three components in the Lebesgue decomposition of the signed IRP measure Π_i: the absolutely continuous part (Proposition 4.5, the existing λ-a.e. condition), the pure-point part (Proposition 4.6, a no-inflation-surprise-premium condition at every event date), and the singular-continuous part (Proposition 4.8 and Theorem 4.9, a Fourier-domain balance of fractal scaling and long-memory structure). Theorem 4.12 translates all three conditions into equalities on the Lévy characteristic triplets (b, σ^2, ν) of the component processes, connecting abstract measure theory to the parameters of standard financial models calibrated from high-frequency data.

Stochastic bank-rate strand. Theorem 8.2 (Section 8) derives a closed-form Itô solution for the bank rate augmented by CIR-type stochastic dynamics, and Corollary 8.3 identifies θ_ξ as the long-run stochastic premium. Theorem 8.5 extends this to a jump-diffusion model consistent with the atomic IRP condition of Proposition 8.7: zero-mean jumps in the stochastic component ensure that scheduled policy events introduce no inflation-risk premium at event dates.

Labour-economics strand. Proposition 9.3 (Section 9) derives an equilibrium labour-fraction formula that embeds the risk-free rate and a labour premium into the labour–leisure decision via a discounting identity, with Corollary 9.4 providing intuitive limiting cases.

Deflation strand. Definition 10.1 (Section 10) introduces the correct notion of deflation as the positive part of negative inflation. Proposition 10.3 and Corollary 10.6 characterise occurrence and the critical tariff threshold; Theorem 10.7 states the full policy-sufficiency condition.

Data-science bridge. Section 11 provides six self-contained algorithms connecting every theoretical construct to a practical workflow: MLE and Gaussian-process regression for parameter estimation; neural-network beta; Q-learning for deflation policy; and, new in this treatise, an event-study algorithm (Algorithm 4) for testing the atomic IRP condition and a multifractal scaling algorithm (Algorithm 5) for testing the singular-continuous IRP condition via the balance relation (57).

The framework is intentionally stylised, and the independence and determinism assumptions of Section 3 are extreme. Relaxing them—allowing stochastic but spanned inflation, or embedding the analysis in a consumption-based model with nominal rigidities—are important avenues for future work. On the measure-theoretic side, the most pressing open question is whether the singular-continuous and pure-point conditions of Theorem 4.3 can be simultaneously satisfied in a *single calibrated model* estimated on joint time series of asset returns and inflation; the event-study and multifractal algorithms of Section 11 provide the statistical machinery for this investigation. Empirical validation of the functional forms and the deflation-prevention conditions of Theorem 10.7 in present-day open economies facing rising tariff barriers is a further priority.

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Glossary

Asset return $r_A(t)$
: Instantaneous expected nominal return on risky asset A at time t. Modelled as a Gaussian probability density in time; equation (18).

Bank rate $r_B(t)$
: Lending rate set by a bank. Modelled here as the zero-IRP deterministic baseline plus a mean-reverting CIR-type stochastic component; equation (27).

CAPM alpha $\alpha(t)$
: Jensen intercept in the extended CAPM equation (24): the portion of the asset return not explained by market-risk exposure.

CAPM beta $\beta_A(t)$
: Sensitivity of asset A's instantaneous return to the market excess return; appears in the CAPM equation (22).

Deflation $d(t)$
: Positive part of negative inflation: $d(t) = [-i(t)]^+$. Deflation is zero when inflation is non-negative and equals $-i(t) > 0$ when the price level falls. Definition 10.1.

Expected inflation
: Conditional expectation $\mathbb{E}_t[i(t)]$ of the instantaneous inflation rate; given explicitly in equation (20).

Fiscal multiplier β_0
: Rate of increase in λ_{eff} per unit of government expenditure G; appears in the prevention condition (45).

Gaussian process (GP)
: Family of random functions in which every finite marginal distribution is Gaussian. Used here for non-parametric inflation regression with Matérn-$\frac{3}{2}$ kernel (50).

Inflation $i(t)$

Instantaneous rate of change of the price-index level. Modelled as $\lambda e^{-\lambda t}\mathbf{1}_{\{t\geq 0\}}$ with $\lambda > 0$ in the normal regime.

Inflation risk premium $p_i(t)$

The component of the nominal expected excess return compensating investors for bearing inflation risk; defined as $p_i = r_A - r_f - \mathbb{E}[i]$. Identically zero in the constructive example; Definitions 3.4 and equation (4).

Itô integral

A stochastic integral with respect to Brownian motion, central to the stochastic bank-rate derivation of Section 8 and Itô's formula (1).

Labour fraction f

Ratio of labour time l to total available time L; governed in equilibrium by equation (40).

Labour premium p_l

Extra return required to compensate an agent for supplying labour, analogous in structure to the inflation risk premium.

MDP

Markov Decision Process: a sequential decision-making framework with states, actions, stochastic transitions, and a reward signal; Definition 11.1.

Matérn kernel

Stationary covariance function for Gaussian processes, parameterised by length-scale ℓ and output variance ς^2; equation (50).

Mean reversion

Property of a stochastic process whereby it is pulled back towards a long-run mean θ_ξ at speed κ, as in the bank-rate SDE (28).

Nominal SDF M_t

Stochastic discount factor (pricing kernel): strictly positive adapted process used to price all nominal payoffs by no-arbitrage; Definition 3.2.

Q-learning

Model-free reinforcement-learning algorithm that estimates the optimal action-value function $Q^*(s,a)$ via temporal-difference updates; Algorithm 3.

Real SDF $\widehat{M_t}$

Nominal SDF deflated by the price index, $\widehat{M_t} = M_t I_t$; prices real payoffs independently of inflation.

Risk-free rate $r_f(t)$

Instantaneous return on a riskless asset; piecewise zero for $t \leq 0$ and a scaled Gaussian-envelope function for $t > 0$; equation (19).

Trade diversion δ

Fraction of exports redirected from tariffed to untariffed markets, reducing effective tariff exposure from τ to $\tau(1-\delta)$ in the prevention condition (45).

Atom (pure-point component)

A time $t_k \in \mathbb{R}$ at which a Borel measure assigns strictly positive mass: $\mu(\{t_k\}) > 0$. Financially, atoms in the IRP measure correspond to scheduled announcement dates at which the expected asset-return jump differs from the combined risk-free-rate and inflation-revision jump; Proposition 4.6.

Borel σ-algebra $\mathcal{B}(\mathbb{R})$

The smallest σ-algebra on $\mathbb{R}$ containing all open sets. The zero-IRP-on-$\mathcal{B}(\mathbb{R})$ condition requires $\Pi_i(B) = 0$ for every element of this σ-algebra, Section 4 and Theorem 4.3.

IRP measure Π_i

The signed Borel measure $\Pi_i = \mu_A - \mu_f - \mu_i$ whose vanishing on all of $\mathcal{B}(\mathbb{R})$ is the subject of Section 4; Definition 4.1.

Lévy measure ν

The jump-intensity measure characterising the frequency and size distribution of jumps in a Lévy process; satisfies $\int(1 \wedge x^2)\nu(\,dx) < \infty$. The Borel zero-IRP condition requires $\nu_A = \nu_f + \nu_i$ on $\mathcal{B}(\mathbb{R} \setminus \{0\})$; Theorem 4.12.

Lebesgue decomposition

The unique decomposition (8) of a signed Borel measure into absolutely continuous (Π_i^{ac}), singular-continuous (Π_i^{sc}), and pure-point (Π_i^{pp}) parts.

Scaling function $\tau(q)$

Function characterising the multifractal structure of a measure via the log-log slope of the q-th moment of cell masses; equation (56). The singular-continuous IRP condition implies the balance $\tau_A(q) = \tau_f(q) + \tau_i(q)$ for all q.

Singular-continuous measure

A Borel measure that is non-atomic yet mutually singular with Lebesgue measure; supported on a Lebesgue-null set. Singular-continuous components of the IRP measure arise from mismatched multifractal or long-memory scaling in returns and inflation; Proposition 4.8 and Remark 4.10.

The End